organisation, so the data doesn't share a physical border with external data. Public platforms cater to many organisations and have logical separation of storage space on a single physical storage entity. Hybrid platforms are also shared like public platforms, but the sharing parties usually belong to the same field of business, which allows them to avail the advantages of a customised architecture that benefits their domain.

Big Data technologies of IoT

The first phase consists of receiving events from IoT connected devices. Wi-Fi, Bluetooth, etc, can be used to connect the devices to receivers. The messages notifying users about events must be sent via an efficient protocol to a broker. MQTT (Message Queue Telemetry Transport) is a popular protocol for transfers among the agents. Mosquitto is a widely used version of a MQTT broker.

In the second phase, upon receiving data, Hadoop and Hive are commonly used to store the data. Apache CouchDB is a NoSQL database which is highly suited for IoT due to its low latency and high throughput. The schema-less database helps with the varying machine data. Apache Storm is preferred for real-time processing and Apache Kafka for intermediate message brokering.

General architecture of IoT Big Data

Context data layer: This collects the external non-IoT data used for IoT data processing later on as extra context/meta data, e.g., start/stop data feeds.

IoT service layer: This handles the interactions between the devices to collect data from IoT devices and also send control commands to them. Bi-directional communication is handled by this layer.

Data/protocol mediator: This is responsible for keeping data in harmonised data entities before it gets published by the data and control layer. This layer is standalone and ensures uniformity.

Data/control broker: This allows third party applications to fire a query or API for accessing harmonised data entities.

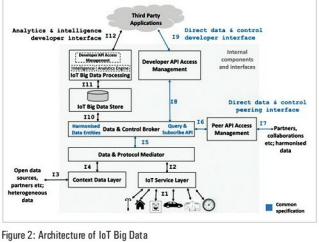


Figure 2: Architecture of IoT Big Data

It also controls requests from the application layer.

Peer API access management: This interacts with peer enterprises to publish relevant context data.

Developer API access management: This controls the permissions for harmonised data entities (both context and IoT) and helps control services provided to third party applications. Access control, authentication and authorisation are managed here. Privacy and security are its main responsibilities.

IoT/Big Data store: This provides short to medium data storage capabilities under the control of the data and control broker. Insights are to be found amongst the ad hoc data relations. Apache Hadoop, Apache Cassandra, MongoDB, etc, are commonly used. Neo4J and Tital are graph databases that are increasingly being used for social media related data.

IoT/Big Data processing: Analytics and business intelligence procedures are carried out here. Analytics includes conventional methods of exploring statistical relationships and the use of analytical engines to produce output though a predefined process. Intelligence signifies usage of artificial intelligence and machine learning to create adaptable algorithms for a match between predicted and desired outcomes. Apache Spark, Apache TinkerPop3, Apache Mahout and TensorFlow are widely used.

Use cases of Big Data and IoT

Fleet management: Many transportation companies carry sensors that monitor drivers' behaviour and a vehicle's location. Good driving skills and on-road safe behaviour get rewarded by insurance companies. IoT gives telematics an advantage by providing detailed machine log data of all the mechanical and electrical components. UPS, the global logistics firm, widely uses this technology to monitor the speed, mileage, break stops, fuel consumption, engine usage, etc, of the vehicles in its fleet. The company hence reduces harmful emissions and fuel consumption.

Healthcare: Wearable fitness tracker and healthcare apps help people monitor their health. Data from these devices can be used to track parameters like blood pressure, sugar levels, etc, as well as to get a prognosis for possible vulnerability to diseases. The Preventice company integrates apps, mobiles, laptops, tablets, the cloud, etc, for remote patient monitoring. The firm allows the customers' doctors to monitor their health online to avoid regular check-ups. Proteus is a startup which has sensors in the pills it makes, which can be used to check if patients are following their prescriptions.

Agriculture: John Deere is a multinational company selling farm equipment. It monitors various parameters like the soil moisture levels, etc. The data goes to a centralised managing platform where, based on the moisture levels, the farmers can be alerted when to irrigate. This prevents unnecessary irrigation and avoids the concentration of water resources in particular areas.